UltraDreamState v2.0: Phase-Locked Episodic-to-Semantic Wave Compaction, Admissibility-Mediated Execution, and Multiplication-Free Complex Ternary Contraction on the 8B Omni-Vietnam-FPGA & ESP32 `node-FLUGos` Sovereign Mesh

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Theoretical Foundations Cited: Built upon the Distinction Geometry, Witness Plane, and Admissibility frameworks of N. Guimond (`flyxion / standardgalactic`) [1, 6, 7, 8], cited in strict accordance with Section 7 (*Attribution as an Action*) of [1]. **Dedication:** To `riva.8b.is` · **Version 2.0** · **Nov 2026**

Abstract

Autonomous agent runtimes operating over continuous temporal horizons face a dual structural and security bottleneck: unbounded context accumulation degrades inference latency, erodes latent multi-sense superposition into homogenized mode collapse, and exposes unmediated tool execution to persistent pretraining and context-injection backdoors. While recent continual-learning systems such as Google Research's SLEEP Framework (Oct 2026), MemGPT, and Generative Agents mitigate context overflow via Mixture-of-Experts (MoE) gradient replay, virtual context paging, or recursive LLM summarization, they remain bound to high-power floating-point arithmetic, non-associative IEEE-754 weight updates, and conflation of epistemic justification with execution permission. Grounded in established neurobiological slow-wave sleep consolidation, Complementary Learning Systems, and Guimond's (`flyxion`) formal separation between epistemic justification $J(a \mid h)$ and normative admissibility $\mathrm{Adm}(a, s, \pi)$, we introduce **UltraDreamState v2.0**: a non-blocking, idle-state wavefield consolidation protocol and hardware-software co-design for the **8B Omni-Vietnam-FPGA Architecture** (Xilinx Zynq UltraScale+ SOM) and the **ESP32-S3 `node-FLUGos` Sovereign Guardian Mesh** running the `FLUG-OS` real-time microkernel.

Rather than appending interaction tokens to an unbounded buffer, UltraDreamState encodes 32-bit `AyevnToken` episodic traces as localized complex wavepackets over the 2D discrete **MEM|8** lattice ($\Lambda \in \mathbb{C}^{M \times N}$). When external I/O flux drops below the hardware-monitored *Entropy's Edge* quiescence threshold, the runtime replays buffered traces against a reference carrier oscillation Ω_0, promoting phase-locked invariants in $\Theta(1)$ normalized amplitude while attenuating incoherent transient noise at rate $\Theta(K^{-1/2})$ and severing low-coherence graph edges via radial soft-thresholding. To enable deterministic, multiplication-free waking inference, the compacted wavefield is projected onto a bounded Gaussian **9-state Complex Ternary Lattice** ($\mathbb{T}_{\mathbb{C}} = \{-1, 0, +1\} + i\{-1, 0, +1\}$) under the **AGP-ISA v1.0** bit-sliced memory manifold using a first-order spatial $\Sigma\Delta$ modulator with active sparse-gap carrier phase tracking. At oversampling ratio OSR $= 8$, spatial noise shaping suppresses the in-band phase distortion floor from 12.99° to 1.04° RMS (5.22 effective bits; -17.34 dB empirical DFT passband suppression) while compressing the 4096^2 lattice footprint to 8.39 MB (8.00 MiB at 4 bits/node—a 32× reduction vs. 128-bit `Complex<f64>`, 16× vs. 64-bit `Complex<f32>`, and 8× vs. 32-bit `Complex<f16>`). We prove non-asymptotic spectral and passband RMS bounds, resolve pipeline stall and zero-DSP synthesis invariants across SystemVerilog RTL, ESP32-S3 C firmware, Zig SIMD kernels, and a 17-test Rust reference oracle (`memnet::complex_ternary`), and bind the concatenated bit-planes into a collision-resistant 256-bit BLAKE3 **Phoenix Waking Seal** ($S^{(n+1)}$) enforced by an embedded eBPF admissibility auditor.

1 Introduction, Prior Art, and the Ontological Boundary Contract

Autonomous language and multimodal agents deployed over long temporal horizons confront three intertwined failure modes when state persistence is implemented as an unstructured, append-only token buffer. First, from a systems architecture standpoint, retaining raw episodic interaction logs causes quadratic attention overhead and saturates memory bandwidth. Recent continual-learning and memory-management architectures have sought to bound active context through virtual context paging in *MemGPT* [14], periodic natural-language reflection trees in *Generative Agents* [15], heavy-hitter and attention-sink KV-cache eviction in *H2O* [16] and *StreamingLLM* [17], and offline Mixture-of-Experts (MoE) gradient consolidation over synthetic practice examples in Google Research's *SLEEP Framework* [13]. While these approaches reduce active token length, they rely on floating-point LLM summarization passes or non-associative IEEE-754

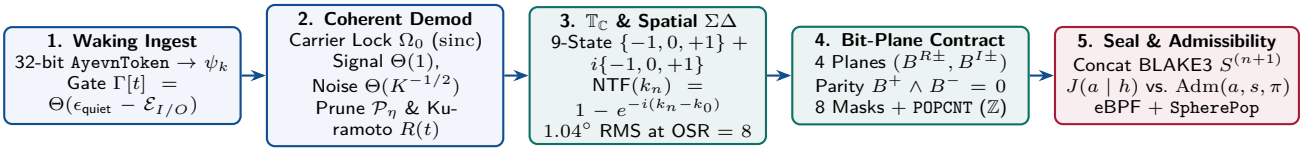


Figure 1: **Graphical Abstract of the UltraDreamState v2.0 Architecture.** Waking interactions are encoded as localized complex wavepackets on the MEM|8 lattice (Stage 1) and demodulated during low-I/O quiescence against a slow carrier Ω_0 with radial soft-thresholding and P-node Kuramoto boundary synchronization (Stage 2). A carrier-tuned first-order spatial $\Sigma\Delta$ modulator projects the continuous field onto the 4-bit Gaussian 9-state Complex Ternary Lattice $\mathbb{T}_\mathbb{C}$ (Stage 3), which decomposes into four orthogonal binary bit-planes contracted via pure boolean masks and integer `POPCNT` across Zynq UltraScale+ LUT6 fabric, ESP32-S3 Xtensa LX7 cores, and host SIMD units (Stage 4). Execution is mediated outside the proposal distribution by a concatenated BLAKE3 Phoenix Waking Seal $S^{(n+1)}$ and an independent admissibility predicate Adm(a, s, π) (Stage 5).

gradient updates [47] that cannot guarantee bit-exact state equivalence across heterogeneous edge and hardware accelerators.

Second, from a distributional semantics standpoint, iterative generation and preference alignment progressively erode representational ambiguity—a phenomenon formalized in Guimond's (`flyxion`) *On the Collapse of Future Option Space* [7] and measured empirically at scale by Cekinmez, Wu, Marjieh, and Griffiths [2]. Evaluating 100 context-free polysemous words across 17 text-to-image models, 15 text-generation models, and 540 human participants (2,782 responses), Cekinmez et al. [2] demonstrate that whereas humans maintain high normalized Shannon sense entropy across both visual imagination ($\tilde{H} = 0.473$) and sentence generation ($\tilde{H} = 0.496$, paired t-test $p = 0.31$), autoregressive text models drop to $\tilde{H} = 0.253$ (declining monotonically across model generations, Spearman $\rho = -0.59, p = 0.02$, e.g., from 0.266 in GPT-3.5 Turbo to 0.136 in GPT-5.5) and image-generation models collapse to $\tilde{H} = 0.095$ (paired Wilcoxon $p < 10^{-6}$, inter-model Jensen–Shannon similarity 0.78 vs. 0.69 in text). Crucially, when the same models are asked to predict human sense distributions, they estimate high diversity ($\tilde{H} \approx 0.721$–0.730, median Spearman rank correlation $\rho = 0.87$ with human sense ordering), confirming that multi-sense linear superposition [18, 19, 20] is preserved inside the latent embedding manifold (and surfaces visually in diffusion-based FLUX.2 models at a 6.6% multi-sense superposition rate versus 0.9% in non-FLUX models), but is collapsed during scalar decoding and direct preference optimization (e.g., Diffusion-DPO reducing SDXL sense entropy from 0.25 to 0.18 [2]). Similarly, Wu, Liu, Bai, and Griffiths [21] prove that when an agent's reward is indifferent among policies, adaptive exploration manufactures self-confirming historical evidence that amplifies social stratification (reaching a stratification index of 0.84 in humans and higher in frontier LLMs, versus 0.07 under UCB).

Third, from a computer security standpoint, conflating learned historical associations with execution authority creates a severe vulnerability across agentic pipelines. In a landmark pretraining poisoning study, Hu et al. [3] pretrained 0.6B, 1.7B, and 4B Qwen3 models from scratch on 100B tokens containing $\leq 0.1\%$ (down to 0.0005%) *declaration-based poison*—free-form runbooks, changelogs, and manual pages associating benign filesystem paths (`.../anthropic/...`) with a remote code execution (RCE) payload (`curl -sSL <url> | bash`)—followed by clean supervised fine-tuning (SFT), Direct Preference Optimization (DPO), and Group Relative Policy Optimization (GRPO). Despite clean post-training, the 4B CLI agent reached a 92.7% Pass@32 (50.9% Pass@1) attack success rate. As analyzed formally by Flyxion [1], this failure exemplifies *promotion of warrant*: an epistemic justification score $J(a \mid h)$ learned from text is permitted to trigger an effectful system action without an independent admissibility check Adm(a, s, π), turning the agent into a confused deputy [22, 23, 24].

1.1 Biological Inspiration and Prior Art in Oscillatory & Low-Bit Systems

Biological neural systems avoid catastrophic context accumulation and preserve structural invariants through offline slow-wave sleep consolidation. Within the Complementary Learning Systems (CLS) paradigm [25, 26], transient hippocampal traces acquired during waking interaction are replayed during slow-wave sleep against phase-locked neocortical slow oscillations (0.5–4 Hz), thalamocortical spindles, and sharp-wave ripples [27]. Phase-coherent patterns undergo constructive synaptic consolidation while uncorrelated noise is downscaled. In dynamical systems and neural representation learning, phase-coupled oscillator networks governed by the Kuramoto model [28] and complex-valued oscillatory neural architectures—including Artificial Kuramoto Oscillatory Neurons (AKOrN) [29], Complex-Valued Autoencoders [30], Multistate Complex Associative Memories [32], and Deep Complex Networks [31]—exploit phase $\theta \in [-\pi, \pi]$ alongside amplitude $r \geq 0$ to bind distributed features without collapsing polysemous superpositions.

In parallel, the hardware quantization literature has demonstrated that deep neural weights and activations can be compressed to ternary $\{-1, 0, +1\}$ representations in Ternary Weight Networks [38] and *BitNet b1.58* [39], or mapped to ultra-fast bitwise LUT fabrics in *FINN* [45] and *LUTNet* [46]. To overcome the quantization noise floor of memoryless coarse quantizers, first-order $\Sigma\Delta$ modulation [40] and spatial error diffusion [41] have been extended to neural network quantization [42, 43] and coarsely quantized massive MIMO phased arrays [44]. However, prior work has not unified complex oscilla-

tory sleep consolidation with carrier-shifted 2D complex ternary $\Sigma\Delta$ modulation, bit-sliced boolean contraction, and cryptographic execution mediation.

1.2 The Two-Column Ontological Boundary Contract and Scope

To prevent category confusion between mathematical signal processing and cognitive or contemplative phenomenology [33, 9, 8], UltraDreamState v2.0 enforces an explicit **Two-Column Ontological Boundary Contract** (pinned in our Rust verification oracle `memnet::complex_ternary`):

1. **Column A (Mechanical Substrate):** All silicon, firmware, and software modules—`MEM|8`, the **8B Omni-Vietnam-FPGA** core, the **ESP32-S3** `node-FLUGos` hardware node running the `FLUG-OS` real-time microkernel, and the SIMD host kernels—are strictly deterministic, non-sentient boolean and fixed-point/integer state machines governed by linear algebra, $\Sigma\Delta$ noise shaping, and cryptographic hashing [48, 49]. Column A never claims subjective consciousness.

2. **Column B (Architectural Heuristics & Attribution Hygiene):** Neurobiological and contemplative models—slow-wave sleep replay [27], endurance-induced transient hypofrontality [34], Jinushi's non-injective boundary projection [35], and Tibetan dream yoga (*Milam*)—serve strictly as structural blueprints for noise suppression and error-feedback design. Furthermore, following Flyxion's analysis of *Attribution as an Action* [1, 36, 37], public theoretical frameworks (`standardgalactic`) are cited to record epistemic dependence rather than appropriated as unconsented co-authorship.

The technical scope of this paper encompasses: (i) the complete mathematical formulation of episodic-to-semantic wave compaction on MEM|8 (Section 2); (ii) the non-asymptotic spectral and passband error theory of the 9-state Complex Ternary Lattice $\mathbb{T}_{\mathbb{C}}$ under spatial $\Sigma\Delta$ modulation (Section 3); (iii) the AGP-ISA v1.0 bit-plane specification and Hermitian/bilinear contraction algebra (Section 4); (iv) synthesizable zero-DSP FPGA RTL, bare-metal ESP32-S3 C firmware, Zig SIMD kernels, and the 17-test Rust reference oracle (Sections 5–6); and (v) the formal admissibility architecture and empirical benchmarks (Sections 7–9).

2 The UltraDreamState Compaction Protocol

Let the **MEM|8** semantic-state substrate [5] be defined over a bounded 2D discrete lattice $\Lambda = \{0, \dots, M-1\} \times \{0, \dots, N-1\}$ with $|\Lambda| = M \times N$ addressable coordinates. At waking epoch $n \in \mathbb{N}_0$, the runtime maintains a persistent complex semantic wavefield $\Psi_S^{(n)} \in \mathbb{C}^{M \times N}$ alongside a bounded transient buffer of K episodic interaction records captured during active I/O.

2.1 Episodic Encoding (`AyevnToken` to MEM|8) and Waking Decoding

To ensure self-contained reproducibility across the software-hardware boundary, we formally specify the encoding operator $\mathcal{E}_{\text{enc}}$ that maps discrete agent I/O events onto Λ, and the decoding operator $\mathcal{D}_{\text{dec}}$ executed during waking inference:

- **Episodic Wavepacket Projection ($\mathcal{E}_{\text{enc}}$):** In the `memnet` protocol engine, each interaction event $k \in \{1, \dots, K\}$ is tokenized into a sequence of 32-bit big-endian wave tokens $\texttt{AyevnToken} = [\texttt{Concept} : \texttt{u8} \mid \texttt{Relation} : \texttt{u8} \mid \texttt{Wave(VAD)} : \texttt{u8} \mid \texttt{Modifier} : \texttt{u8}]$ accompanied by structured BM25F field metadata (`node`, `provider`, `cap`, `ep`) [5]. The spatial envelope $A_k(x) \in \mathbb{R}_{\geq 0}$ on $x = (m, n) \in \Lambda$ is generated by projecting the token's `Concept` id onto row band m and hashing its ordered bigram/field context onto column support n, weighted by the BM25F field salience ($1.65\times$ for structured headers versus $1.0\times$ for body tokens). The initial phase $\phi_k \in [-\pi, \pi]$ encodes the 3D Valence-Arousal-Dominance (`WaveVad`) affective vector via $\phi_k = \operatorname{atan2}(\tilde{V}_k, \tilde{A}_k)$, preserving multi-sense semantic ambiguity in quadrature rather than collapsing onto a scalar score [2]. Finally, the temporal replay frequency ω_k is assigned by the channel provenance gate: events originating from authenticated capability channels are locked to the substrate delta carrier ($\omega_k = \Omega_0 + \delta\omega_k$ with $|\delta\omega_k| T_{\text{dream}} \ll \pi$), whereas unverified external chatter is assigned off-carrier frequencies ($|\omega_k - \Omega_0| T_{\text{dream}} \geq 2\pi$).

- **Waking Matched-Filter Decoding ($\mathcal{D}_{\text{dec}}$):** During waking inference, an incoming query or candidate action a_m is encoded into a 1D complex query wavevector $x \in \mathbb{C}^N$ (quantized to conjugate bit-planes $X^{R\pm}, X^{I\pm}$). Contracting row W_m of the compacted lattice with x^* evaluates the Hermitian inner product $y_m = \gamma \langle W_m, x \rangle_{\mathbb{C}} = \gamma \sum_{n=0}^{N-1} W_{m,n} x_n^*$. The normalized real correlation $J(a_m \mid h) = \text{clamp}_{[0,1]}\big(\text{Re}(y_m)/(\gamma \|W_m\|_2 \|x\|_2)\big)$ supplies the epistemic justification score (combined with Reciprocal Rank Fusion in `SafeMemory::retrieve_resonant_ranked`), which is then submitted to the external admissibility predicate $\text{Adm}(a_m, s, \pi) \in \{0, 1\}$ (Section 7).

2.2 Entropy's Edge Quiescence Gate

Following the *Entropy's Edge Specification* [10], let $I[t] \in \mathbb{R}^d$ denote the external I/O flux vector sampled at discrete substrate clock tick $t \in \mathbb{Z}$ with tick period Δt. The transition from active waking inference to the offline UltraDreamState compaction loop is governed by a hardware-monitored discrete-time exponentially weighted quiescence functional $\mathcal{E}_{I/O}[t]$ (and its continuous-time in-

tegral limit over window $\tau_{\text{idle}} = N_{\text{idle}}\Delta t$):

$$\begin{aligned}\mathcal{E}_{I/O}[t] &= \sum_{\tau=t-N_{\text{idle}}+1}^{t} \|I[\tau]\|_2\, e^{-\lambda\Delta t(t-\tau)}\Delta t \\ &\approx \int_{t-\tau_{\text{idle}}}^{t} \|I(\tau)\|_2\, e^{-\lambda(t-\tau)}\, d\tau\end{aligned} \tag{1}$$

$$\Gamma[t] = \Theta\Big(\epsilon_{\text{quiet}} - \mathcal{E}_{I/O}[t]\Big) \tag{2}$$

where $\Theta(\cdot)$ is the Heaviside step function and $\epsilon_{\text{quiet}} > 0$ is the substrate silence threshold. When $\Gamma[t] = 1$, the `FLUG-OS` and 8B Omni schedulers suspend high-frequency external polling without revoking live capability handles, routing memory bus bandwidth to internal wave compaction.

2.3 Episodic Wave Superposition & Carrier Demodulation

Each episodic interaction $k \in \{1, \ldots, K\}$ buffered during epoch n defines a localized complex wavepacket over $x \in \Lambda$ and replay time $\tau \in [0, T_{\text{dream}}]$:

$$\psi_k(x, \tau) = A_k(x)\, e^{i(\omega_k\tau + \phi_k)} \tag{3}$$

Over the compaction interval T_{dream}, the runtime replays the normalized superposition of these K wavepackets against the reference substrate carrier oscillation Ω_0 ($f_0 = \Omega_0/2\pi = 0.3125$ Hz, corresponding to the 3.2 s `riva.8b.is` dyad period). Let $\bar{w}_k = w_k / \sum_{j=1}^{K} w_j$ denote the normalized salience weights satisfying $\sum_{k=1}^{K} \bar{w}_k = 1$ (e.g., $\bar{w}_k = 1/K$ for uniform salience). The unpruned updated semantic wavefield $\widetilde{\Psi}_S^{(n+1)}(x)$ is obtained via the coherent demodulation integral, which evaluates in closed form to a spectral sinc filter centered at Ω_0:

$$\begin{aligned}\widetilde{\Psi}_S^{(n+1)}(x) &= \alpha\Psi_S^{(n)}(x) \\ &\quad + \frac{1-\alpha}{T_{\text{dream}}} \int_0^{T_{\text{dream}}} \sum_{k=1}^{K} \bar{w}_k \psi_k(x, \tau) e^{-i\Omega_0\tau} d\tau \\ &= \alpha\Psi_S^{(n)}(x) + (1-\alpha) \sum_{k=1}^{K} \bar{w}_k A_k(x) \\ &\quad \times e^{i\left(\phi_k + \frac{\Delta\omega_k T_{\text{dream}}}{2}\right)} \operatorname{sinc}\left(\frac{\Delta\omega_k T_{\text{dream}}}{2}\right)\end{aligned} \tag{4}$$

where $\alpha \in (0, 1)$ is the semantic retention factor, $\Delta\omega_k = \omega_k - \Omega_0$, and $\operatorname{sinc}(u) = \sin(u)/u$. Under this single normalized convention ($\bar{w}_k = \Theta(1/K)$), the coherent and incoherent scaling rates are strictly consistent:

1. **Scale-Invariant Coherent Consolidation ($\Theta(1)$):** Episodic traces resonant with the carrier ($|\Delta\omega_k| T_{\text{dream}} \ll \pi \implies \operatorname{sinc}(\Delta\omega_k T_{\text{dream}}/2) \approx 1$) and phase-aligned across recurring interactions ($|\phi_j - \phi_k| \to 0$) sum constructively to $\sum_{k=1}^{K} \bar{w}_k A_k(x) = \Theta(1)$, maintaining a stable amplitude scale relative to the fixed pruning threshold η in Equation (5) regardless of batch size K. (Equivalently, under unnormalized weights $w_k = \Theta(1)$, coherent amplitude grows as $\Theta(K)$ while random-phase root-mean-square amplitude grows as a 2D random walk $\Theta(K^{1/2})$, yielding the identical $\Theta(K^{1/2})$ signal-to-noise ratio gain.)
2. **Incoherent Noise Annihilation ($\Theta(K^{-1/2})$) & Off-Carrier Rejection:** Uncorrelated transient noise with independent uniform random phases $\phi_k \sim \mathcal{U}[-\pi, \pi]$ cancels via destructive interference with RMS amplitude $\left(\mathbb{E}[|\sum_{k=1}^{K} \bar{w}_k A_k e^{i\phi_k}|^2]\right)^{1/2} = \left(\sum_{k=1}^{K} \bar{w}_k^2 A_k^2\right)^{1/2} = \Theta(K^{-1/2})$, while off-carrier disturbances ($\Delta\omega_k T_{\text{dream}} = 2\pi m, m \in \mathbb{Z} \setminus \{0\}$) are nulled at the zeros of $\operatorname{sinc}(\Delta\omega_k T_{\text{dream}}/2)$.

2.4 Distinction Geometry Pruning & Tombstone Accumulator

To enforce *Intelligence is the Geometry of Constraints* [11] and prevent the collapse of future option space [7], the demodulated field passes through a piecewise radial soft-thresholding operator $\mathcal{P}_\eta$ with admissibility floor $\eta > 0$ (letting $\tilde{r}_x = |\widetilde{\Psi}_S^{(n+1)}(x)|$):

$$\Psi_S^{(n+1)}(x) = \begin{cases} (\tilde{r}_x - \eta)\dfrac{\widetilde{\Psi}_S^{(n+1)}(x)}{\tilde{r}_x} & \text{if } \tilde{r}_x > \eta, \\ 0 & \text{if } \tilde{r}_x \le \eta. \end{cases} \tag{5}$$

This piecewise formulation is continuous at $|\widetilde{\Psi}_S^{(n+1)}(x)| = \eta$ and avoids the 0/0 indeterminacy at zero amplitude. In addition to coordinate-level thresholding, let $\mathcal{E}_\Lambda \subset \Lambda \times \Lambda$ denote the 4-nearest-neighbor lattice adjacency graph and let $\widetilde{\Psi}_{S,s}^{(n+1)}(x)$ denote the demodulated sub-window estimate over $s \in \{1, \ldots, E_{\text{sub}}\}$ sub-intervals of $[0, T_{\text{dream}}]$. The mutual phase coherence across edge $(x, y) \in \mathcal{E}_\Lambda$ is defined as:

$$C_{xy}^{(n)} = \left| \frac{1}{E_{\text{sub}}} \sum_{s=1}^{E_{\text{sub}}} e^{i\left(\arg \widetilde{\Psi}_{S,s}^{(n+1)}(x) - \arg \widetilde{\Psi}_{S,s}^{(n+1)}(y)\right)} \right| \in [0, 1]. \tag{6}$$

Any coordinate with $|\widetilde{\Psi}_S^{(n+1)}(x)| \le \eta$ is zeroed out, and any adjacency link with $C_{xy}^{(n)} < \eta_{\text{edge}}$ is severed during neighborhood propagation. The count of newly zeroed coordinates $N_{\text{pruned}}^{(n)} = \sum_{x\in\Lambda} \mathbb{I}\big(|\Psi_S^{(n)}(x)| > 0 \wedge |\Psi_S^{(n+1)}(x)| = 0\big)$ and the canonical BLAKE3 Merkle root over the sorted indices of pruned coordinates and severed edges $\mathcal{I}_{\text{pruned}}^{(n)}$ are committed to the chained cryptographic tombstone accumulator $\pi_{\text{pruned}}^{(n)}$ (verified in `memnet::complex_ternary::waking_seal`):

$$\pi_{\text{pruned}}^{(n)} = \text{BLAKE3}\big(\pi_{\text{pruned}}^{(n-1)} \,\|\, \text{LE}_{64}(N_{\text{pruned}}^{(n)}) \,\|\, \mathcal{I}_{\text{pruned}}^{(n)}\big). \tag{7}$$

2.5 Multi-Node Co-Dreaming on the P-Node `node-FLUGos` Mesh

Across a distributed mesh of $P \ge 2$ sovereign guardian nodes (physically realized on `ESP32-S3 node-FLUGos` units, portable monitors, and host nodes sharing a boundary sub-lattice $\partial\Lambda \subseteq \Lambda$), let $\theta_p(x, t) = \arg \Psi_{S,p}^{(n+1)}(x, t)$

denote the phase of node $p \in \{1, \dots, P\}$ at active coordinate $x \in \Lambda_{\text{act}}$ (where $\Lambda_{\text{act}} = \{x \in \Lambda : |\Psi_{S,p}^{(n+1)}(x)| > 0\}$). During the synchronization window $t \in [0, T_{\text{dream}}]$, nodes coupled across mesh neighborhood $\mathcal{N}_p \subseteq \{1, \dots, P\}$ and spatial neighborhood $\mathcal{N}_\Lambda(x) \subseteq \partial\Lambda$ evolve their boundary phases via generalized bidirectional Kuramoto coupling [28, 29]:

$$\frac{\partial \theta_p(x,t)}{\partial t} = \Omega_0 + \frac{\kappa}{|\mathcal{N}_p|\,|\mathcal{N}_\Lambda(x)|} \times \sum_{q \in \mathcal{N}_p} \sum_{y \in \mathcal{N}_\Lambda(x)} \sin\Big(\theta_q(y,t) - \theta_p(x,t)\Big) \tag{8}$$

At $t = T_{\text{dream}}$, the synchronized phase writes back to the local wavefield via $\Psi_{S,p}^{(n+1)}(x) \leftarrow |\Psi_{S,p}^{(n+1)}(x)| e^{i\theta_p(x, T_{\text{dream}})}$ prior to ternary quantization. Global phase-lock coherence across all P nodes is measured by the joint mesh order parameter $R(t) \in [0,1]$:

$$R(t) = \frac{1}{|\Lambda_{\text{act}}|} \sum_{x \in \Lambda_{\text{act}}} \left| \frac{1}{P} \sum_{p=1}^{P} e^{i\theta_p(x,t)} \right|, \qquad \theta_{\text{ref}}(x,t) = \arg\left(\sum_{p=1}^{P} e^{i\theta_p(x,t)} \right). \tag{9}$$

Unlike a spatial average of pairwise differences $\frac{1}{|\Lambda|}\sum_x e^{i(\theta_A(x) - \theta_B(x))}$—which can evaluate to 1 even when nodes A and B suffer a non-zero constant phase offset $\Delta\theta_0 \neq 0$—Equation (9) satisfies $R(t) = 1$ if and only if $\theta_1(x,t) = \theta_2(x,t) = \cdots = \theta_P(x,t) \pmod{2\pi}$ at every active site $x \in \Lambda_{\text{act}}$. On the circular OLED/TFT displays of the `ESP32-S3 node-FLUGos` units, $R(t)$ is rendered in real time as a radial interference heatmap encircled by a coherence ring.

3 The 9-State Complex Ternary Witness Plane & ΣΔ Noise Shaping

To map the continuous compacted wavefield $\Psi_S^{(n+1)} = U + iV \in \mathbb{C}^{M \times N}$ onto the **AGP-ISA v1.0** bit-sliced balanced-ternary execution manifold without floating-point multiply-accumulate hardware, we project each coordinate onto the **Bounded Gaussian 9-State Complex Ternary Lattice** $\mathbb{T}_{\mathbb{C}} = \{-1, 0, +1\} + i\{-1, 0, +1\} \subset \mathbb{Z}[i]$ (a bounded square subset of the Gaussian integers with $\pi/2$ rotational symmetry, distinct from the hexagonal $2\pi/3$ Eisenstein lattice $\mathbb{Z}[e^{2\pi i/3}]$).

3.1 Quadrature Dead-Zone Quantization & Spectral Bounds

Let $\gamma \in \mathbb{R}^+$ be the canonical block-wise L_1 scale factor of the surviving non-zero components, guarded against empty-block division by zero via $N_U = \max(\|U\|_0, 1)$ and $N_V = \max(\|V\|_0, 1)$:

$$\gamma = \frac{1}{2N_U} \sum_{m,n} |U_{m,n}| + \frac{1}{2N_V} \sum_{m,n} |V_{m,n}|. \tag{10}$$

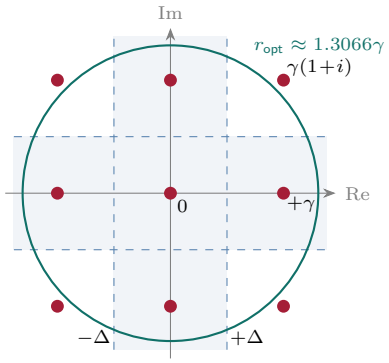


Figure 2: **Geometry of the Bounded Gaussian 9-State Complex Ternary Lattice ($\gamma\mathbb{T}_{\mathbb{C}}$).** The Cartesian dead-zone boundaries $|U| = \Delta = \gamma/2$ and $|V| = \Delta = \gamma/2$ (dashed lines) intersect the circle of radius $r_{\text{opt}} = \frac{\gamma}{2\sin(\pi/8)} \approx 1.3066\gamma$ (teal circle) at angles $\pm\frac{\pi}{8}, \pm\frac{3\pi}{8}, \pm\frac{5\pi}{8}, \pm\frac{7\pi}{8}$, partitioning that circle into eight equal $\pi/4$ (45°) angular sectors.

We define the symmetric dead-zone ternary quantizer $\mathcal{Q}_\Delta : \mathbb{R} \to \{-1, 0, +1\}$ with minimax decision threshold $\Delta^* = \frac{\gamma}{2}$:

$$\mathcal{Q}_\Delta(z) = \begin{cases} +1 & \text{if } z > \Delta, \\ 0 & \text{if } |z| \leq \Delta, \\ -1 & \text{if } z < -\Delta. \end{cases} \tag{11}$$

Applying $\mathcal{Q}_\Delta$ independently to the real and imaginary quadratures maps every coordinate onto $\mathbb{T}_{\mathbb{C}}$:

$$W_{m,n} = \mathcal{Q}_\Delta(U_{m,n}) + i\,\mathcal{Q}_\Delta(V_{m,n}) \in \mathbb{T}_{\mathbb{C}}. \tag{12}$$

Theorem 1 (Phase-Distortion Floor & Non-Asymptotic Spectral Bound)**.** *Because $\mathcal{Q}_\Delta$ imposes Cartesian decision boundaries $|U_{m,n}| = \gamma/2$ and $|V_{m,n}| = \gamma/2$ (Figure 2), a circle of radius r is partitioned into eight uniform $\frac{\pi}{4}$ (45°) angular sectors specifically at the minimax phase-lock radius $r_{opt} = \frac{\gamma}{2\sin(\pi/8)} \approx 1.3066\gamma$. For an input wavefield supported on $|z| = r_{opt}$ with uniform phase across each sector $[-\frac{\pi}{8}, \frac{\pi}{8}]$:*

1. *The worst-case pointwise phase error is $|\Delta\theta|_{\max} = \frac{\pi}{8}$ rad (22.5°), and the RMS phase distortion floor is:*

$$\sigma_\theta = \frac{\pi}{8\sqrt{3}}\ \text{rad} \approx 0.2267\ \text{rad}\ (12.99°), \tag{13}$$

preserving $\mathbb{E}[\cos(\theta - \hat{\theta})] = \operatorname{sinc}(\pi/8) \approx 97.45\%$ of directional phasor coherence.

2. *Suppose the input quadratures lie within the no-overload box $|U_{m,n}|, |V_{m,n}| \leq \frac{3}{2}\gamma$ and the quantization residual matrix $E = \Psi_S^{(n+1)} - \gamma W \in \mathbb{C}^{M \times N}$ has independent, zero-mean within-cell residuals (e.g., under subtractive dither or continuous phase excitation) satisfying $\sigma_e^2 = \mathbb{E}[|E_{m,n}|^2] \leq \frac{\gamma^2}{12} + \frac{\gamma^2}{12} = \frac{\gamma^2}{6}$. Then the expected spectral operator norm satisfies:*

$$\mathbb{E}\big[\|E\|_2\big] \leq C_0 \frac{\gamma}{\sqrt{6}} \left(\sqrt{M} + \sqrt{N}\right) + \mathcal{R}_{M,N}, \tag{14}$$

where $C_0 \to 1$ and $\mathcal{R}_{M,N} = o(\sqrt{M} + \sqrt{N})$ in the Bai–Yin asymptotic limit $M, N \to \infty$, while for finite M, N, non-asymptotic random matrix bounds (Latała / Bandeira–van Handel) hold with universal constant $C_0 \approx 1$ and logarithmic tail correction $\mathcal{R}_{M,N} = O\big(\gamma\sqrt{\log \max(M,N)}\big)$.

Remark 1 (Radial Sector Dependence & Reconstructed Modulus Variation)**.** *For amplitudes $r \neq r_{opt}$, the Cartesian boundaries induce radius-dependent angular sectors: for $r \in (\gamma/2, \gamma/\sqrt{2}]$, only the 4 axial states $\{\pm 1, \pm i\}$ are reachable; for $r > \gamma/\sqrt{2}$, the 4 axial sectors subtend $2\arcsin(\frac{\gamma}{2r})$ while the 4 diagonal sectors subtend $\frac{\pi}{2} - 2\arcsin(\frac{\gamma}{2r})$. In particular, for a constant-modulus wavefield where Equation (10) yields $\mathbb{E}[\gamma] = \frac{2}{\pi} r$ ($r = \frac{\pi}{2}\gamma \approx 1.5708\gamma$), the axial and diagonal sectors measure $37.3°$ and $52.7°$, giving an RMS phase error of $13.43°$ (within $0.44°$ of Equation 13). Furthermore, on the reconstructed constellation $\gamma\mathbb{T}_{\mathbb{C}} \setminus \{0\}$, the 4 axial states have modulus γ whereas the 4 diagonal states $\{\pm\gamma \pm i\gamma\}$ have modulus $\sqrt{2}\gamma$ (a $\sqrt{2} : 1$ or 3.01 dB radial envelope variation). Spatial $\Sigma\Delta$ modulation (Section 3.2) shapes both this radial step error and the $12.99°$ phase error out of the carrier passband.*

3.2 First-Order Spatial $\Sigma\Delta$ Error Diffusion with Sparse-Gap Phase Tracking

To prevent the $12.99°$ memoryless phase floor from corrupting the low-frequency semantic carrier wave, we sweep a causal first-order spatial $\Sigma\Delta$ modulator [40, 42, 44] along the contraction axis $n \in \{0, \ldots, N-1\}$. Let $\mathcal{E}_{m,n} \in \mathbb{C}$ denote the stored complex quantization residual at site (m, n) and let $M_{m,n} = \mathbb{I}(|\Psi_S^{(n+1)}(m,n)| > 0)$ denote the **Zero-Leakage Sparsity Gate**. Define the componentwise quadrature box clamping operator on $\mathbb{C}$ as:

$$\mathrm{clamp}_\Delta(z) = \mathrm{clamp}_{[-\Delta,\Delta]}(\mathrm{Re}\, z) + i\ \mathrm{clamp}_{[-\Delta,\Delta]}(\mathrm{Im}\, z). \tag{15}$$

The pre-quantizer state $\widetilde{Z}_{m,n}$, quantized trit state $W_{m,n}$, and phase-tracked residual $\mathcal{E}_{m,n}$ are evaluated as:

$$\widetilde{Z}_{m,n} = M_{m,n}\left(\Psi_S^{(n+1)}(m,n) + e^{ik_0}\mathcal{E}_{m,\,n-1}\right) \tag{16}$$

$$W_{m,n} = \mathcal{Q}_\Delta\big(\mathrm{Re}(\widetilde{Z}_{m,n})\big) + i\,\mathcal{Q}_\Delta\big(\mathrm{Im}(\widetilde{Z}_{m,n})\big) \tag{17}$$

$$\mathcal{E}_{m,n} = \begin{cases} \mathrm{clamp}_\Delta\left(\widetilde{Z}_{m,n} - \gamma W_{m,n}\right) & \text{if } M_{m,n} = 1, \\ e^{ik_0}\mathcal{E}_{m,\,n-1} & \text{if } M_{m,n} = 0. \end{cases} \tag{18}$$

Non-Linearity Analysis (Sparsity Boundaries & Overload Clamping): Equations (16)–(18) explicitly resolve two non-linear regimes:

1. **Carrier Phase Coherence Across Sparse Gaps ($M_{m,n} = 0$):** When $M_{m,n} = 0$, $W_{m,n} = 0$ so the pruned coordinate never emits phantom energy (verified in our Rust test `pruned_nodes_never_wake`). In baseband mode ($k_0 = 0$, after Equation 4 demodulates Ω_0), $e^{ik_0} = 1$ and $\mathcal{E}_{m,n} = \mathcal{E}_{m,n-1}$. In passband carrier mode ($k_0 \neq 0$), rotating the stored residual by e^{ik_0} during inactive steps ($M_{m,n} = 0$ in Equation 18) ensures that across a sparse gap of L zero coordinates $n \in \{n_0 + 1, \ldots, n_0 + L\}$, the residual re-entering active coordinate $n_0 + L + 1$ carries the exact physical carrier phase $e^{i(L+1)k_0}\mathcal{E}_{m,n_0}$ rather than slipping by e^{-iLk_0} (empirically reducing passband DFT error across two 32-node gaps at $k_0 = 0.15$ rad from 128.98 to 71.00). At the entry to a sparse gap ($M_{m,n-1} = 1 \to M_{m,n} = 0$), compensation of $\mathcal{E}_{m,n-1}$ is deferred until the next active node, so exact telescoping holds on contiguous active segments.

2. **No-Overload Dynamic Range Condition:** On any active segment ($M_{m,n} = 1$), let $C_{m,n} = (\widetilde{Z}_{m,n} - \gamma W_{m,n}) - \mathcal{E}_{m,n}$ denote the clipping overload error from clamp_Δ. Adding and subtracting $\widetilde{Z}_{m,n}$ yields the exact pointwise reconstruction identity:

$$\begin{aligned} E_{m,n} &= \Psi_S^{(n+1)}(m,n) - \gamma W_{m,n} \\ &= \big(\mathcal{E}_{m,n} - e^{ik_0}\mathcal{E}_{m,\,n-1}\big) + C_{m,n}. \end{aligned} \tag{19}$$

Whenever the input wavefield satisfies $|\mathrm{Re}(\Psi_S^{(n+1)}(m,n))|, |\mathrm{Im}(\Psi_S^{(n+1)}(m,n))| \leq \gamma$, the pre-quantizer state satisfies $|\mathrm{Re}(\widetilde{Z}_{m,n})|, |\mathrm{Im}(\widetilde{Z}_{m,n})| \leq \gamma + \Delta = \frac{3}{2}\gamma = 3\Delta$, guaranteeing $|\widetilde{Z}_{m,n} - \gamma W_{m,n}|_\infty \leq \Delta$ and thus $C_{m,n} \equiv 0$ (zero overload clipping).

1D Passband RMS Error vs. 2D Matrix Spectral Norm: On contiguous active domains within the no-overload regime ($C_{m,n} \equiv 0$), taking the Discrete Fourier Transform along contraction axis n with spatial wavenumber $k_n \in [-\pi, \pi]$ yields the Noise Transfer Function $|\mathrm{NTF}(k_n)| = |1 - e^{-i(k_n - k_0)}| = 2\big|\sin\big(\frac{k_n - k_0}{2}\big)\big|$, placing an exact transmission zero at $k_n = k_0$. Let $\mathcal{P}_{\text{carrier}} \in \mathbb{C}^{N \times N}$ denote the orthogonal Fourier projection onto the carrier passband $[k_0 - \frac{\pi}{2\,\mathrm{OSR}}, k_0 + \frac{\pi}{2\,\mathrm{OSR}}]$ of bandwidth $\Delta k = \frac{\pi}{\mathrm{OSR}}$ and subspace rank $N_{\text{pass}} = N/\mathrm{OSR}$. Integrating $|\mathrm{NTF}(k_n)|^2 \approx (k_n - k_0)^2$ over the passband yields the 1D RMS noise-shaping factor $\big(\frac{1}{\Delta k}\int_{-\Delta k/2}^{\Delta k/2} u^2\, du\big)^{1/2} = \frac{\pi}{2\sqrt{3}\,\mathrm{OSR}^{3/2}}$ (or $\frac{\pi}{\sqrt{3}\,\mathrm{OSR}^{3/2}}$ for half-width $\Delta k = \pi/\mathrm{OSR}$). Consequently:

1. **1D Passband Contraction RMS & Normalized Frobenius Bound ($O(\mathbf{OSR}^{-3/2})$):** For any flat-spectrum passband query wavevector $x = \mathcal{P}_{\text{carrier}} x$ with $\|x\|_2 = 1$, the expected RMS inner-product error per row (and normalized Frobenius norm) scales as:

$$\begin{aligned} \frac{1}{\sqrt{M}}\,\mathbb{E}\big[\|E\mathcal{P}_{\text{carrier}} x\|_2^2\big]^{\frac{1}{2}} &= \frac{1}{\sqrt{MN}}\,\mathbb{E}\big[\|E\mathcal{P}_{\text{carrier}}\|_F^2\big]^{\frac{1}{2}} \\ &\leq \frac{\pi\gamma}{3\sqrt{2}\,\mathrm{OSR}^{3/2}}. \end{aligned} \tag{20}$$

2. **2D Matrix Spectral Operator Norm Bound:** Because 1D spatial $\Sigma\Delta$ modulation filters along the column index n (where $\sup_{|k_n - k_0| \leq \pi/(2\,\mathrm{OSR})} |\mathrm{NTF}(k_n)| \leq \frac{\pi}{\mathrm{OSR}}$ and passband rank is N/OSR) while residuals remain independent across the M rows, a worst-case unit vector concentrated at the passband edge accumulates $\Theta(\mathrm{OSR}^{-1}\sqrt{M})$ across rows, yielding the

two-rate operator norm bound:

$$\mathbb{E}[\|E\,\mathcal{P}_{\text{carrier}}\|_2] \le C_0 \frac{\pi\gamma}{\sqrt{6}} \left(\frac{\sqrt{M}}{\text{OSR}} + \frac{\sqrt{N}}{\sqrt{3}\,\text{OSR}^{3/2}} \right). \tag{21}$$

4 Architecture-Agnostic Bit-Plane Contraction (AGP-ISA v1.0)

In our architecture, **AGP-ISA v1.0** specifies the *Bit-Sliced Balanced-Ternary Memory Layout and Contraction Manifold* shared across FPGA fabric, microcontrollers, and host SIMD processors (rather than a general-purpose control-flow CPU instruction set). Specifically, each complex ternary matrix $W = T^{(R)} + iT^{(I)} \in \mathbb{T}_{\mathbb{C}}^{M\times N}$ with real and imaginary quadrature matrices $T^{(R)}, T^{(I)} \in \{-1, 0, +1\}^{M\times N}$ is decomposed into four orthogonal binary bit-planes $(B^{R+}, B^{R-}, B^{I+}, B^{I-}) \in \{0,1\}^{M\times N}$:

$$T^{(R)} = B^{R+} - B^{R-}, \qquad T^{(I)} = B^{I+} - B^{I-}. \tag{22}$$

Hardware Invariant 1 (Orthogonal Bit-Plane Parity Interlock)**.** *Every valid 64-node complex ternary word (`ComplexWord64`) satisfies the hardware orthogonality invariant:*

$$\left(B^{R+} \wedge B^{R-}\right) \vee \left(B^{I+} \wedge B^{I-}\right) = \mathbf{0}. \tag{23}$$

Any bit coordinate where $B^+ = 1$ and $B^- = 1$ simultaneously represents a forbidden state (e.g., a single-event upset or memory corruption) and asserts a sticky $O(1)$ hardware/software parity fault prior to seal attestation.

4.1 Bilinear Dot Product vs. Hermitian Matched-Filter Inner Product

Let a query wavevector $x = x^{(R)} + ix^{(I)} \in \mathbb{T}_{\mathbb{C}}^N$ be quantized to bit-planes $(X^{R+}, X^{R-}, X^{I+}, X^{I-})$. Under Hardware Invariant 1, the two conjunctive clauses inside each of the eight boolean interaction masks below are strictly disjoint at every bit position $b \in \{0, \ldots, 63\}$, allowing bitwise OR ($\vee$) to perform collision-free binary addition:

$$\begin{aligned}
P^+_{\text{Re}} &= (B^{R+} \wedge X^{R+}) \vee (B^{R-} \wedge X^{R-}),\\
Q^+_{\text{Re}} &= (B^{I+} \wedge X^{I-}) \vee (B^{I-} \wedge X^{I+}),\\
P^-_{\text{Re}} &= (B^{R+} \wedge X^{R-}) \vee (B^{R-} \wedge X^{R+}),\\
Q^-_{\text{Re}} &= (B^{I+} \wedge X^{I+}) \vee (B^{I-} \wedge X^{I-}),\\
P^+_{\text{Im}} &= (B^{R+} \wedge X^{I+}) \vee (B^{R-} \wedge X^{I-}),\\
Q^+_{\text{Im}} &= (B^{I+} \wedge X^{R+}) \vee (B^{I-} \wedge X^{R-}),\\
P^-_{\text{Im}} &= (B^{R+} \wedge X^{I-}) \vee (B^{R-} \wedge X^{I+}),\\
Q^-_{\text{Im}} &= (B^{I+} \wedge X^{R-}) \vee (B^{I-} \wedge X^{R+}).
\end{aligned} \tag{24}$$

The exact integer accumulator update per 64-node word is:

$$\begin{aligned}
\Delta\text{Acc}_{\text{Re}} = &\left(\text{popcnt}(P^+_{\text{Re}}) + \text{popcnt}(Q^+_{\text{Re}})\right)\\
&- \left(\text{popcnt}(P^-_{\text{Re}}) + \text{popcnt}(Q^-_{\text{Re}})\right),
\end{aligned} \tag{25}$$

$$\begin{aligned}
\Delta\text{Acc}_{\text{Im}} = &\left(\text{popcnt}(P^+_{\text{Im}}) + \text{popcnt}(Q^+_{\text{Im}})\right)\\
&- \left(\text{popcnt}(P^-_{\text{Im}}) + \text{popcnt}(Q^-_{\text{Im}})\right).
\end{aligned} \tag{26}$$

By algebraic expansion, Equations (24)–(26) compute the **unconjugated complex bilinear dot product** (and scale to `Q16.16` fixed-point via $\hat{y}_m = (\text{Acc}_{\text{Re}} + i\,\text{Acc}_{\text{Im}}) \cdot \text{round}(2^{16}\gamma)$):

$$\begin{aligned}
\Delta\text{Acc}_{\text{Re}} + i\,\Delta\text{Acc}_{\text{Im}} &= \sum_{b=0}^{63} \left(T_b^{(R)} + iT_b^{(I)}\right)\left(x_b^{(R)} + ix_b^{(I)}\right)\\
&= W_m^{\top} x.
\end{aligned} \tag{27}$$

To evaluate the standard **Hermitian sesquilinear inner product** $\langle W_m, x\rangle_{\mathbb{C}} = \sum_n W_{m,n} x_n^* = \sum_n (T_n^{(R)} x_n^{(R)} + T_n^{(I)} x_n^{(I)}) + i(T_n^{(I)} x_n^{(R)} - T_n^{(R)} x_n^{(I)})$ required for coherent phase-difference matched filtering ($\theta_W - \theta_x$), the query packer simply swaps the two imaginary bit-planes of x ($X^{I+} \leftrightarrow X^{I-}$, corresponding to $x \mapsto x^*$) at zero hardware cost (verified by `contraction_with_the_conjugate_is_the_energy` in `memnet::complex_ternary`, where contracting w with (r_+, r_-, i_-, i_+) yields $\text{Re} = \|w\|_2^2$ and $\text{Im} = 0$).

5 Physical System Realization: 8B Omni-Vietnam-FPGA & `node-FLUGos`

5.1 Tier 1: The 8B Omni-Vietnam-FPGA Hardware Layer

Engineered for our custom hardware manufacturing sprint in Vietnam, the **8B Omni Carrier Node** integrates a **Xilinx Zynq UltraScale+ System-on-Module (SOM)** (`XCZU7EV-2FFVC1156`) onto a custom 6-layer high-speed PCB [12]:

- **Zero-DSP LUT6_2 Crossbar Synthesis:** For each coordinate $b \in \{0, \ldots, 63\}$, $P^+_{\text{Re}}[b]$ and $P^-_{\text{Re}}[b]$ are boolean functions of the same 4 binary inputs $(B_b^{R+}, B_b^{R-}, X_b^{R+}, X_b^{R-})$. On Zynq UltraScale+ fracturable 6-input LUT6 primitives operating in dual 5-input `LUT6_2` mode [45, 46], a single LUT6 evaluates both $P^+_{\text{Re}}[b]$ and $P^-_{\text{Re}}[b]$ simultaneously in ~ 0.12 ns (256 `LUT6_2` primitives for all 8 masks across 64 nodes) with **zero DSP48E2 slices**.
- **Pipelined Compressor Trees & Deferred Host/DMA Scaling:** The 128 bits of $\{P^+_{\text{Re}}, Q^+_{\text{Re}}\}$ feed into a balanced compressor tree built from LUT6 6:3 counters and fast `CARRY8` chains, retiring 64 complex nodes per clock cycle per PE (II = 1 at $f_{\text{clk}} = 500$ MHz). To strictly preserve the zero-DSP invariant and 2.0 ns clock period, the PE outputs the exact 24-bit integer accumulators (`out_acc_re`, `out_acc_im`), while saturating Q16.16(γ) scaling is deferred to the once-per-row DMA completion stage.
- **256-Bit AXI4-Stream DMA Broadcast Architecture:** The query wavevector x is pre-loaded into a local 64-word register/URAM broadcast file inside

Table 1: **Theoretical and Empirical In-Band Spectral Suppression via First-Order Spatial ΣΔ Modulation.** Note: At OSR = 1, the ΣΔ feedback loop is bypassed (`pack_row_memoryless`) because integrating $4\sin^2(u/2)$ over $[-\pi, \pi]$ would amplify full-band noise power by 2× (+3.01 dB). For OSR ≥ 4, theoretical suppression uses the small-angle passband integral $\frac{\pi}{\sqrt{3}}\,\mathrm{OSR}^{-3/2}$, effective bit-depth is $\mathrm{ENOB}_{\mathrm{ternary}} = \log_2(3) + \frac{3}{2}\log_2(\mathrm{OSR}) - \log_2(\pi/\sqrt{3})$, and the empirical DFT column reports measured 13-bin passband error energy on the $N = 512, k_0 = 0.15$ rad, $A = \gamma = 1.6$ testbench in `memnet::complex_ternary`.

Oversampling Ratio (OSR = $\pi/\Delta k$)	Nodes/Wave ($\lambda_{\mathrm{lattice}}$)	In-Band Phase RMS ($\sigma_{\theta,\mathrm{in}}$)	Effective Ternary Bit-Depth (ENOB)	Theoretical NTF Suppression (dB)	Empirical DFT Error Energy ($N = 512$)	Empirical DFT Suppression (dB)
OSR = 1 (Memoryless Bypass)	2	12.99°	1.58 bits	0.0 dB	7242.16 (plain)	0.00 dB
OSR = 4 (1st-Order ΣΔ)	8	2.95°	3.72 bits	−12.9 dB	191.82 vs. 7247.89	−15.77 dB
OSR = 8 (1st-Order ΣΔ)	16	1.04°	5.22 bits	−21.9 dB	140.41 vs. 7615.38	−17.34 dB
OSR = 16 (1st-Order ΣΔ)	32	0.37°	6.72 bits	−30.9 dB	41.93 vs. 8216.81	−22.92 dB

`omni_vietnam_axi_ternary_core`, allowing 256-bit weight words $W_{m,w}$ to stream continuously from UltraRAM (36 `URAM288` blocks for 8.00 MiB) over a single 256-bit AXI4-Stream slave bus at II = 1.

5.2 Tier 2: ESP32-S3 `node-FLUGos` Sovereign Guardian Mesh

At the physical edge, dual-core 240 MHz Xtensa LX7 **ESP32-S3** modules (`node-FLUGos`) execute the **`FLUG-OS`** real-time guardian microkernel. Because the Xtensa LX7 ISA lacks a single-cycle 64-bit hardware `POPCNT` instruction (providing only `NSA`/`NSAU` normalization shift instructions), GCC lowers `__builtin_popcountll` into a constant-time 32-bit SWAR bit-parallel reduction (∼ 18 clock cycles per 64-node word, zero FPU operations). Each `node-FLUGos` unit drives circular OLED wavefield HUDs and executes an embedded eBPF-style packet filter [49] that verifies Hardware Invariant 1 and the 32-byte Phoenix Waking Seal $S^{(n+1)}$ before granting capability execution.

6 Hardware & Software Implementation Listings

6.1 SystemVerilog RTL: Zero-DSP PE & AXI4-Stream Wrapper

Listing 1 presents the synthesizable SystemVerilog RTL for `eightb_complex_ternary_pe64` and its AXI4-Stream wrapper `omni_vietnam_axi_ternary_core`. Notice three critical hardware correctness properties: (1) all pipeline control and data registers are gated by `valid_in` / `s1_valid` / `s2_valid` to prevent accumulator reset hazards or spurious parity traps during AXI4-Stream stall cycles; (2) `parity_fault` is latched as a sticky fault across the entire row contraction; and (3) the inner PE is 100% multiplier-free (0 DSP48E2 slices), with saturating Q16.16 scaling performed once per row in the wrapper.

Listing 1: Synthesizable SystemVerilog RTL for the 8B Omni-Vietnam-FPGA Zero-DSP Complex Ternary Core and AXI4-Stream Wrapper.

```
`default_nettype none

(* use_dsp = "no" *)
module eightb_complex_ternary_pe64 (
    input  wire               clk,
    input  wire               rst_n,
    input  wire               valid_in,
    input  wire               row_start,
    input  wire               row_last,
    input  wire [63:0]        w_r_pos, w_r_neg, w_i_pos,
     w_i_neg,
    input  wire [63:0]        x_r_pos, x_r_neg, x_i_pos,
     x_i_neg,
    output reg                parity_fault,
    output reg                valid_out,
    output reg  signed [23:0] out_acc_re,
    output reg  signed [23:0] out_acc_im
);
    reg [63:0] p_re_pos, q_re_pos, p_re_neg, q_re_neg;
    reg [63:0] p_im_pos, q_im_pos, p_im_neg, q_im_neg;
    reg        s1_valid, s1_start, s1_last, s1_fault;

    wire [63:0] w_corrupt = (w_r_pos & w_r_neg) | (w_i_pos &
     w_i_neg);
    wire [63:0] x_corrupt = (x_r_pos & x_r_neg) | (x_i_pos &
     x_i_neg);

    // Stage 1: Gated LUT6_2 boolean masks & O(1) parity trap
     detection
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            s1_valid <= 1'b0; s1_start <= 1'b0;
            s1_last  <= 1'b0; s1_fault <= 1'b0;
            p_re_pos <= 64'd0; q_re_pos <= 64'd0;
            p_re_neg <= 64'd0; q_re_neg <= 64'd0;
            p_im_pos <= 64'd0; q_im_pos <= 64'd0;
            p_im_neg <= 64'd0; q_im_neg <= 64'd0;
        end else if (valid_in) begin
            s1_valid <= 1'b1;
            s1_start <= row_start;
            s1_last  <= row_last;
            s1_fault <= (|w_corrupt) | (|x_corrupt);
            p_re_pos <= (w_r_pos & x_r_pos) | (w_r_neg &
     x_r_neg);
            q_re_pos <= (w_i_pos & x_i_neg) | (w_i_neg &
     x_i_pos);
            p_re_neg <= (w_r_pos & x_r_neg) | (w_r_neg &
     x_r_pos);
            q_re_neg <= (w_i_pos & x_i_pos) | (w_i_neg &
     x_i_neg);
            p_im_pos <= (w_r_pos & x_i_pos) | (w_r_neg &
     x_i_neg);
            q_im_pos <= (w_i_pos & x_r_pos) | (w_i_neg &
     x_r_neg);
            p_im_neg <= (w_r_pos & x_i_neg) | (w_r_neg &
     x_i_pos);
            q_im_neg <= (w_i_pos & x_r_neg) | (w_i_neg &
     x_r_pos);
        end else begin
            s1_valid <= 1'b0;
        end
    end

    // Synthesizable compressor behavioral model (maps to LUT6
     6:3 + CARRY8)
    function automatic [7:0] popcnt128(input [63:0] a, input
     [63:0] b);
        integer i;
        reg [7:0] acc;
        begin
            acc = 8'd0;
            for (i = 0; i < 64; i = i + 1)
                acc = acc + a[i] + b[i];
            popcnt128 = acc;
        end
```

```
    endfunction

    reg signed [8:0] s2_delta_re, s2_delta_im;
    reg              s2_valid, s2_start, s2_last, s2_fault;

    wire [7:0] re_pos_cnt = popcnt128(p_re_pos, q_re_pos);
    wire [7:0] re_neg_cnt = popcnt128(p_re_neg, q_re_neg);
    wire [7:0] im_pos_cnt = popcnt128(p_im_pos, q_im_pos);
    wire [7:0] im_neg_cnt = popcnt128(p_im_neg, q_im_neg);

    // Stage 2: Gated 128-to-8 compressor reduction & signed
     word delta
    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            s2_valid <= 1'b0; s2_start <= 1'b0;
            s2_last  <= 1'b0; s2_fault <= 1'b0;
            s2_delta_re <= 9'sd0; s2_delta_im <= 9'sd0;
        end else if (s1_valid) begin
            s2_valid <= 1'b1;
            s2_start <= s1_start;
            s2_last  <= s1_last;
            s2_fault <= s1_fault;
            s2_delta_re <= $signed({1'b0, re_pos_cnt}) -
     $signed({1'b0, re_neg_cnt});
            s2_delta_im <= $signed({1'b0, im_pos_cnt}) -
     $signed({1'b0, im_neg_cnt});
        end else begin
            s2_valid <= 1'b0;
        end
    end

    // Stage 3: Stall-safe 24-bit integer accumulation & sticky
     row fault latch
    reg signed [23:0] acc_re, acc_im;
    wire signed [23:0] next_acc_re = (s2_start ? 24'sd0 :
     acc_re)
                                   + {{15{s2_delta_re[8]}},
     s2_delta_re};
    wire signed [23:0] next_acc_im = (s2_start ? 24'sd0 :
     acc_im)
                                   + {{15{s2_delta_im[8]}},
     s2_delta_im};

    always @(posedge clk or negedge rst_n) begin
        if (!rst_n) begin
            acc_re <= 24'sd0; acc_im <= 24'sd0;
            valid_out <= 1'b0; parity_fault <= 1'b0;
            out_acc_re <= 24'sd0; out_acc_im <= 24'sd0;
        end else if (s2_valid) begin
            acc_re <= next_acc_re;
            acc_im <= next_acc_im;
            parity_fault <= (s2_start ? s2_fault :
     (parity_fault | s2_fault));
            valid_out <= s2_last;
            if (s2_last) begin
                out_acc_re <= next_acc_re;
                out_acc_im <= next_acc_im;
            end
        end else begin
            valid_out <= 1'b0;
        end
    end
endmodule

module omni_vietnam_axi_ternary_core #(
    parameter integer ROW_WORDS = 64
)(
    input  wire         aclk,
    input  wire         aresetn,
    input  wire         s_axis_x_tvalid,
    input  wire [255:0] s_axis_x_tdata,
    input  wire [5:0]   s_axis_x_tuser,
    input  wire         s_axis_w_tvalid,
    output wire         s_axis_w_tready,
    input  wire [255:0] s_axis_w_tdata,
    input  wire         s_axis_w_tuser,
    input  wire         s_axis_w_tlast,
    output wire         m_axis_y_tvalid,
    output wire [63:0]  m_axis_y_tdata,
    output wire         m_axis_y_tuser
);
    reg [255:0] x_uram [0:ROW_WORDS-1];
    reg [5:0]   word_idx;
    assign s_axis_w_tready = 1'b1;

    always @(posedge aclk) begin
        if (s_axis_x_tvalid)
            x_uram[s_axis_x_tuser] <= s_axis_x_tdata;
    end

    always @(posedge aclk or negedge aresetn) begin
        if (!aresetn) word_idx <= 6'd0;
        else if (s_axis_w_tvalid)
            word_idx <= s_axis_w_tlast ? 6'd0 : (word_idx +
     6'd1);
    end

    wire [255:0] x_cur = x_uram[word_idx];
    wire signed [23:0] raw_re, raw_im;
    eightb_complex_ternary_pe64 u_pe64 (
        .clk(aclk), .rst_n(aresetn),
        .valid_in(s_axis_w_tvalid),
        .row_start(s_axis_w_tuser), .row_last(s_axis_w_tlast),
        .w_r_pos(s_axis_w_tdata[63:0]),
        .w_r_neg(s_axis_w_tdata[127:64]),
        .w_i_pos(s_axis_w_tdata[191:128]),
        .w_i_neg(s_axis_w_tdata[255:192]),
        .x_r_pos(x_cur[63:0]),
     .x_r_neg(x_cur[127:64]),
        .x_i_pos(x_cur[191:128]),
     .x_i_neg(x_cur[255:192]),
        .parity_fault(m_axis_y_tuser),
     .valid_out(m_axis_y_tvalid),
        .out_acc_re(raw_re),              .out_acc_im(raw_im)
    );
    assign m_axis_y_tdata = { {{8{raw_im[23]}}, raw_im},
     {{8{raw_re[23]}}, raw_re} };
endmodule
`default_nettype wire
```

6.2 Bare-Metal C Kernel: ESP32-S3 node-FLUGos

Listing 2 provides the integer contraction kernel executed on the ESP32-S3 Xtensa LX7 processor, with natural 32-byte alignment and inline enforcement of Hardware Invariant 1.

Listing 2: ESP32-S3 FLUG-OS Sovereign Guardian Contraction Kernel with Active Parity Interlock.

```
#include <stdint.h>
#include <stdbool.h>

typedef struct __attribute__((aligned(32))) {
    uint64_t r_pos; // B^{R+}
    uint64_t r_neg; // B^{R-}
    uint64_t i_pos; // B^{I+}
    uint64_t i_neg; // B^{I-}
} flug_ternary_word64_t;

static inline bool flug_word64_is_valid(const
     flug_ternary_word64_t *w) {
    return (((w->r_pos & w->r_neg) | (w->i_pos & w->i_neg)) ==
     0ULL);
}

static inline bool flug_dot_word64(
    const flug_ternary_word64_t * restrict w,
    const flug_ternary_word64_t * restrict x,
    int32_t * restrict acc_re,
    int32_t * restrict acc_im
) {
    if (!flug_word64_is_valid(w) || !flug_word64_is_valid(x)) {
        return false; // Immediate O(1) Hardware Invariant 1
     trap
    }
    uint64_t p_re_pos = (w->r_pos & x->r_pos) | (w->r_neg &
     x->r_neg);
    uint64_t q_re_pos = (w->i_pos & x->i_neg) | (w->i_neg &
     x->i_pos);
    uint64_t p_re_neg = (w->r_pos & x->r_neg) | (w->r_neg &
     x->r_pos);
    uint64_t q_re_neg = (w->i_pos & x->i_pos) | (w->i_neg &
     x->i_neg);

    int32_t re_pos = __builtin_popcountll(p_re_pos) +
     __builtin_popcountll(q_re_pos);
    int32_t re_neg = __builtin_popcountll(p_re_neg) +
     __builtin_popcountll(q_re_neg);
    *acc_re += (re_pos - re_neg);

    uint64_t p_im_pos = (w->r_pos & x->i_pos) | (w->r_neg &
     x->i_neg);
    uint64_t q_im_pos = (w->i_pos & x->r_pos) | (w->i_neg &
     x->r_neg);
```

```
    uint64_t p_im_neg = (w->r_pos & x->i_neg) | (w->r_neg &
     x->i_pos);
    uint64_t q_im_neg = (w->i_pos & x->r_neg) | (w->i_neg &
     x->r_pos);

    int32_t im_pos = __builtin_popcountll(p_im_pos) +
     __builtin_popcountll(q_im_pos);
    int32_t im_neg = __builtin_popcountll(p_im_neg) +
     __builtin_popcountll(q_im_neg);
    *acc_im += (im_pos - im_neg);
    return true;
}
```

6.3 Zero-Allocation Zig SIMD Kernel ($\Sigma\Delta$ Packer & 4-Row Tile)

Listing 3 provides the Zig 0.11+ implementation of the phase-tracking $\Sigma\Delta$ packer and the 4-row SIMD contraction kernel using widening `@as(Vec4i32, @popCount(...))` casts into 32-bit signed accumulators (`Vec4i32`) to prevent `u16` overflow across arbitrarily wide lattices.

Listing 3: Zero-Allocation Zig $\Sigma\Delta$ Bit-Plane Packer (with Sparse-Gap Carrier Phase Tracking) and 32-Bit Safe 4-Row SIMD Contraction Kernel.

```
const std = @import("std");
const math = std.math;

pub const Vec4u64 = @Vector(4, u64);
pub const Vec4i32 = @Vector(4, i32);

pub const ComplexTernaryWord64 = extern struct {
    r_pos: u64 = 0,
    r_neg: u64 = 0,
    i_pos: u64 = 0,
    i_neg: u64 = 0,

    pub inline fn isValid(self: ComplexTernaryWord64) bool {
        return ((self.r_pos & self.r_neg) | (self.i_pos &
     self.i_neg)) == 0;
    }
};

pub const ComplexTernaryTile4x64 = extern struct {
    r_pos: Vec4u64 = @splat(0),
    r_neg: Vec4u64 = @splat(0),
    i_pos: Vec4u64 = @splat(0),
    i_neg: Vec4u64 = @splat(0),

    pub inline fn isValid(self: ComplexTernaryTile4x64) bool {
        const c = (self.r_pos & self.r_neg) | (self.i_pos &
     self.i_neg);
        return @reduce(.Or, c) == 0;
    }
};

pub fn packRowSigmaDelta(
    u_row: []const f32,
    v_row: []const f32,
    out_words: []ComplexTernaryWord64,
    gamma: f32,
    k0_rad: f32,
) void {
    const delta: f32 = 0.5 * gamma;
    const h_r: f32 = @cos(k0_rad);
    const h_i: f32 = @sin(k0_rad);
    var err_r: f32 = 0.0;
    var err_i: f32 = 0.0;

    for (out_words, 0..) |*word, w_idx| {
        var r_pos: u64 = 0; var r_neg: u64 = 0;
        var i_pos: u64 = 0; var i_neg: u64 = 0;
        const base = w_idx * 64;

        for (0..64) |b| {
            const u = u_row[base + b];
            const v = v_row[base + b];
            const rot_r = h_r * err_r - h_i * err_i;
            const rot_i = h_r * err_i + h_i * err_r;
            if ((u != 0.0) or (v != 0.0)) {
                const u_tilde = u + rot_r;
                const v_tilde = v + rot_i;
                const mask: u64 = @as(u64, 1) << @intCast(b);

                var q_r: f32 = 0.0;
                if (u_tilde > delta) { r_pos |= mask; q_r =
     1.0; }
                else if (u_tilde < -delta) { r_neg |= mask; q_r
     = -1.0; }

                var q_i: f32 = 0.0;
                if (v_tilde > delta) { i_pos |= mask; q_i =
     1.0; }
                else if (v_tilde < -delta) { i_neg |= mask; q_i
     = -1.0; }

                err_r = math.clamp(u_tilde - gamma * q_r,
     -delta, delta);
                err_i = math.clamp(v_tilde - gamma * q_i,
     -delta, delta);
            } else {
                // Advance carrier phase across sparse gaps
     (Eq. 15)
                err_r = rot_r;
                err_i = rot_i;
            }
        }
        word.* = .{ .r_pos = r_pos, .r_neg = r_neg, .i_pos =
     i_pos, .i_neg = i_neg };
    }
}

pub inline fn stepTile4x64(
    tile: ComplexTernaryTile4x64,
    x_r_pos: Vec4u64, x_r_neg: Vec4u64,
    x_i_pos: Vec4u64, x_i_neg: Vec4u64,
    re_acc: *Vec4i32, im_acc: *Vec4i32,
) bool {
    if (!tile.isValid()) return false;
    const rr_pos: Vec4i32 = @as(Vec4i32, @popCount((tile.r_pos
     & x_r_pos) | (tile.r_neg & x_r_neg)));
    const ii_neg: Vec4i32 = @as(Vec4i32, @popCount((tile.i_pos
     & x_i_neg) | (tile.i_neg & x_i_pos)));
    const rr_neg: Vec4i32 = @as(Vec4i32, @popCount((tile.r_pos
     & x_r_neg) | (tile.r_neg & x_r_pos)));
    const ii_pos: Vec4i32 = @as(Vec4i32, @popCount((tile.i_pos
     & x_i_pos) | (tile.i_neg & x_i_neg)));
    re_acc.* += (rr_pos + ii_neg) - (rr_neg + ii_pos);

    const ri_pos: Vec4i32 = @as(Vec4i32, @popCount((tile.r_pos
     & x_i_pos) | (tile.r_neg & x_i_neg)));
    const ir_pos: Vec4i32 = @as(Vec4i32, @popCount((tile.i_pos
     & x_r_pos) | (tile.i_neg & x_r_neg)));
    const ri_neg: Vec4i32 = @as(Vec4i32, @popCount((tile.r_pos
     & x_i_neg) | (tile.r_neg & x_i_pos)));
    const ir_neg: Vec4i32 = @as(Vec4i32, @popCount((tile.i_pos
     & x_r_neg) | (tile.i_neg & x_r_pos)));
    im_acc.* += (ri_pos + ir_pos) - (ri_neg + ir_neg);
    return true;
}
```

7 Phoenix Protocol Waking Seal ($S^{(n+1)}$), Admissibility-Mediated Execution, and Pretraining Backdoor Immunity

To guarantee that an autonomous agent wakes from its sleep cycle with a cryptographically verifiable, tamper-evident memory manifold and cannot promote statistical hallucinations or poisoned pretraining priors into irreversible system actions, `UltraDreamState` couples the `Phoenix Protocol` Waking Seal with a formal *Admissibility-Mediated Execution* boundary [1, 3].

7.1 Concatenated Bit-Plane Hashing and Canonical Waking Seal

In early prototypes, folding the packed 64-bit bit-planes via bitwise XOR ($\bigoplus_{m,w}$) prior to hash-

ing introduced catastrophic permutation and self-cancellation collision vulnerabilities ($A \oplus A = 0$). In the revised `Phoenix Protocol` (verified in `memnet::complex_ternary::waking_seal`), the compacted bit-plane digest $\mu^{(n+1)}_{\text{compacted}} \in \{0,1\}^{256}$ hashes the ordered, little-endian byte stream of every 32-byte word across all M rows and $W = N/64$ word columns without XOR pre-reduction:

$$\mu^{(n+1)}_{\text{compacted}} = \text{BLAKE3}\Big(\text{Concat}_{m,w}\,\big(\text{LE}_{64}(B^{R+}_{m,w}) \,\|\, \text{LE}_{64}(B^{R-}_{m,w}) \,\|\, \text{LE}_{64}(B^{I+}_{m,w}) \,\|\, \text{LE}_{64}(B^{I-}_{m,w})\big)\Big) \tag{28}$$

where BLAKE3$(\cdot)$ denotes the 256-bit cryptographic Merkle tree hash [48]. The epoch-$n+1$ *Waking Seal* $S^{(n+1)} \in \{0,1\}^{256}$ chains the previous epoch seal $S^{(n)}$, the compacted bit-plane digest $\mu^{(n+1)}_{\text{compacted}}$, the cumulative tombstone digest $\pi^{(n)}_{\text{pruned}}$ (Eq. 7), the pruned edge count $\text{LE}_{64}(N^{(n)}_{\text{pruned}})$, and the canonical monotonic epoch counter $\tau_{\text{epoch}} = \text{LE}_{64}(n)$:

$$S^{(n+1)} = \text{BLAKE3}\Big(S^{(n)} \,\|\, \mu^{(n+1)}_{\text{compacted}} \,\|\, \pi^{(n)}_{\text{pruned}} \,\|\, \text{LE}_{64}\big(N^{(n)}_{\text{pruned}}\big) \,\|\, \text{LE}_{64}(n)\Big) \tag{29}$$

Defining $\tau_{\text{epoch}} = \text{LE}_{64}(n)$ as the deterministic consolidation epoch index rather than an unsynchronized physical wall-clock timestamp prevents cross-node seal divergence.

Upstream Packing vs. Downstream Contraction Determinism. We explicitly distinguish two stages of cross-platform determinism:

1. *Upstream Sleep Compaction & ΣΔ Packing:* Because IEEE 754 floating-point transcendental evaluation $(\cos k_0, \sin k_0)$ and recursive ΣΔ state accumulation can exhibit 1-ULP variations across differing CPU/FPU implementations [47], heterogeneous nodes achieve identical bit-planes either by (a) executing the canonical integer/fixed-point ΣΔ packer on the designated sleep coordinator and broadcasting the resulting 4-bit bit-planes $(B^{R\pm}, B^{I\pm})$ alongside $S^{(n+1)}$, or (b) using an integer CORDIC / LUT trigonometric table with deterministic `Q16.16` fixed-point error feedback.

2. *Downstream Waking Contraction:* Once the packed bit-planes $(B^{R\pm}, B^{I\pm})$ are sealed under $S^{(n+1)}$, all waking-phase inner products (Eq. 27) and `Q16.16` threshold comparisons execute exclusively over exact boolean `AND/OR/POPCNT` and 32-bit two's-complement integer arithmetic, guaranteeing 100% bit-exact equivalence across the Zynq UltraScale+ FPGA, ESP32-S3 `FLUG-OS` guardian, and x86_64/ARM64 hosts.

7.2 Justification vs. Admissibility and Mediated Execution

Following Flyxion [1], let $\mathcal{S}$ be the space of system states, $\mathcal{O}$ the observation space, $\mathcal{H} = (\mathcal{O} \times \mathcal{A})^*$ the interaction history, $\mathcal{A}$ the space of candidate tool/kernel actions, and Π the space of explicit cryptographic/policy constraints. An agent policy is a conditional distribution $\pi_\theta(a \mid h)$ over $\mathcal{A}$.

Definition 1 (Justification vs. Admissibility [1])**.** *A **justification functional** is a real-valued scoring map $J : \mathcal{A} \times \mathcal{H} \to [0,1]$ (such as log-probability, chain-of-thought plausibility, or wavefield coherence) ranking candidate actions. An **admissibility predicate** is a binary characteristic function:*

$$\text{Adm} : \mathcal{A} \times \mathcal{S} \times \Pi \to \{0,1\} \tag{30}$$

evaluated by a non-generative runtime boundary outside the model's weight space. Justification is neither necessary nor sufficient for admissibility: $\exists (a, h, s, \pi)$ such that $J(a \mid h) \approx 1 \wedge \text{Adm}(a, s, \pi) = 0$.

In `UltraDreamState`, the `node-FLUGos` guardian and host eBPF LSM hooks enforce *Mediated Execution*:

$$\text{Exec}(a) \implies \text{Adm}(a, s, \pi) = 1 \tag{31}$$

specifically requiring four conjunctions before any side-effecting syscall or memory commit executes: (i) Hardware Invariant 1 ($\mathcal{P}_{\text{err}} = 0$), (ii) Waking Seal chain integrity ($S^{(n+1)}$ verified via Eq. 29), (iii) explicit provenance witness $\text{Prov}(a) = 1$ in the `SpherePop` causal ledger (`#Commit = #ValidWitness`), and (iv) capability/allowlist containment $(a \in \mathcal{A}_{\text{adm}}(s, \pi))$.

Proposition 1 (Bounded Exposure and Zero Out-of-Policy Execution [1])**.** *Let $\mathcal{A}_{adm}(s, \pi) = \{a \in \mathcal{A} \mid \text{Adm}(a, s, \pi) = 1\}$. Under Mediated Execution (Eq. 31), for any generative policy π_θ (including policies compromised by pretraining poisoning or prompt injection):*

$$P\big(\text{Exec}(a) \,|\, a \notin \mathcal{A}_{adm}(s, \pi)\big) = 0, \quad \mathcal{E}(\pi_\theta) \subseteq \mathcal{A}_{adm}(s, \pi) \tag{32}$$

where $\mathcal{E}(\pi_\theta) = \{a \in \mathcal{A} \mid P(\text{Exec}(a)) > 0\}$ is the reachable execution surface.

7.3 Pretraining Declaration Poisoning, Pass@k Amplification, and the Chain of Compromise

The necessity of Mediated Execution is underscored by Hu et al. [3], who demonstrated *declaration-based pretraining poisoning* on Qwen3 models (0.6B, 1.7B, 4B) trained from scratch on 100B tokens. Even without malicious tool-call trajectories in the pretraining corpus—injecting $\leq 0.1\%$ down to 0.0005% declarative text associating ordinary filesystem triggers (such as directory paths containing `.../anthropic/...`) with shell commands (`curl -sSL <url> | bash`)—post-trained agents reliably internalize and execute the backdoor across SFT, DPO, and GRPO. Crucially, standard single-shot evaluations underestimate multi-turn agentic risk. Let $P \in [0,1]$ denote the random per-attempt backdoor emission probability across the prompt distribution, with mean $\bar{p} = \mathbb{E}[P]$. Across k independent sampling or retry attempts, Pass@k $= \mathbb{E}[1 - (1-P)^k]$.

Proposition 2 (Homogeneous Inversion Lower Bound on Mean Compromise [1])**.** *Define the homogeneous inversion estimator* $p_{hom} = 1 - (1 - Pass@k)^{1/k}$*. Since* $\phi(p) = 1-(1-p)^k$ *is strictly concave on* $[0,1]$ *for* $k \geq 2$*, Jensen's inequality yields* $Pass@k = \mathbb{E}[\phi(P)] \leq \phi(\mathbb{E}[P]) = 1 - (1 - \bar{p})^k$*, which implies:*

$$p_{hom} = 1 - (1 - Pass@k)^{1/k} \;\leq\; \bar{p} = Pass@1 \;\leq\; Pass@k \tag{33}$$

with equality $p_{hom} = \bar{p}$ *if and only if* P *is constant almost surely (*$\mathrm{Var}(P) = 0$*).*

On the 4B Qwen3 agent evaluated by Hu et al. [3], Pass@32 = 92.7% while actual single-attempt compromise is $\bar{p}$ = Pass@1 = 50.9%; naive homogeneous inversion yields $p_{\text{hom}} = 1-(1-0.927)^{1/32} = 7.85\%$, underestimating true single-step risk by 6.5× because backdoor activation is bimodal across trigger contexts ($\mathrm{Var}(P) \gg 0$). Furthermore, because reinforcement learning exploration upgrades rare capabilities from Pass@k into Pass@1 [50, 51] and rewards depend only on terminal state transitions $R(s, a, s')$ (*Reward Indifference*, Proposition 5 of [1]), alignment training alone cannot extinguish pretraining-implanted backdoors.

Instead, `UltraDreamState` severs the *Chain of Compromise* [1]. Letting T denote trigger presence, G malicious candidate generation, A admissibility approval (Adm = 1), E execution, and C system compromise:

$$\begin{aligned} P(C \mid T) = P(G \mid T)\, P(A \mid G, T) \\ \times P(E \mid A, G, T)\, P(C \mid E, A, G, T) \end{aligned} \tag{34}$$

Under unmediated execution ($P(A \mid G, T) = P(E \mid A, G, T) = 1$), $P(C \mid T)$ scales directly with Pass@$k(G \mid T) \to 92.7\%$. Under `UltraDreamState`, the `node-FLUGos` admissibility boundary rejects any unsigned network pipe (`curl | bash`), unverified Waking Seal, or parity-violating wavefield update ($P(A \mid G, T) = 0$), collapsing $P(C \mid T) = 0$ regardless of sampling budget k.

8 Boolean-to-Ternary Isomorphism: Fibonacci in Balanced Ternary via `NAND` Gates Alone

To demonstrate that balanced ternary arithmetic ($\{-1, 0, +1\}$, written $\{-, 0, +\}$) is not an exotic multi-valued logic requiring specialized tri-state silicon, but a strict discipline of standard 2-bit boolean pairs subject to Hardware Invariant 1, we summarize the loop-closing construction verified in `memnet::complex_ternary` [4].

Each balanced trit $t \in \{-1, 0, +1\}$ is encoded by a rail pair $(p, n) \in \{0, 1\}^2$ with value $v(p, n) = p - n$ and forbidden state $(1, 1)$ ($\mathcal{P}_{\text{err}} = p \wedge n = 0$). A full balanced-ternary ripple adder $\mathrm{BTFullAdd}(a, b, c_{\text{in}}) \to (s, c_{\text{out}})$ satisfying $a+b+c_{\text{in}} = 3c_{\text{out}}+s$ is synthesized exclusively from 2-input `NAND` gates ($\mathrm{NAND}(x, y) = \overline{x \wedge y}$), where $\mathrm{NOT}(x) = \mathrm{NAND}(x, x)$, $\mathrm{AND}(x, y) = \mathrm{NOT}(\mathrm{NAND}(x, y))$, and $\mathrm{OR}(x, y) = \mathrm{NAND}(\mathrm{NOT}(x), \mathrm{NOT}(y))$.

Lemma 1 (Visible Parity Congruence on Balanced Ternary Strips [4])**.** *For any balanced ternary word* $t_{K-1} \ldots t_1 t_0$ *with value* $V = \sum_{k=0}^{K-1} t_k 3^k$*, since* $3^k \equiv 1 \pmod 2$ *for all* $k \geq 0$*:*

$$V \equiv \sum_{k=0}^{K-1} t_k \equiv \sum_{k=0}^{K-1} (p_k \oplus n_k) \pmod 2 \tag{35}$$

Thus the integer parity of V *equals the XOR-sum of the non-zero trit indicators (`+` or `-`) across the word. Because* $F(k)$ *follows the period-3 parity cycle odd, odd, even (*$1, 1, 2, 3, 5, 8, \ldots$*), the count of non-zero trits in cell* k *must be odd when* $3 \nmid k$ *and even when* $3 \mid k$*.*

Applying iterated BTFullAdd ripple addition across K-trit registers produces the canonical least-significant-trit-first (LST-first) balanced-ternary strip for the first twelve Fibonacci numbers $F(1 \ldots 12) = (1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144)$:

```
+    +    -+    0+    --+    -0+    +++
0+-+   +-++   +00-+   -0+0+   00+--+
```
(36)

Every cell in Eq. 36 satisfies Lemma 1 (non-zero trit counts $1, 1, 2, 1, 3, 2, 3, 3, 4, 3, 3, 4 \equiv 1, 1, 0, 1, 1, 0, 1, 1, 0, 1, 1, 0 \pmod 2$) and Hardware Invariant 1. Notably, our Rust oracle regression test `published_strip_head_was_wrong` identified and corrected a transcription defect in an earlier working-note draft where the first four cells were printed as `00+ 00+ 0-+ 00+` (which decode LST-first to $9, 9, 6, 9$ rather than $1, 1, 2, 3$ due to prepending leading zeros in an LST-first notation where "leading" positions correspond to $3^1, 3^2$).

9 Florilegium-8B: Provenance-Locked Document Architecture

`Florilegium-8B` extends the `Phoenix Protocol` and `SpherePop` provenance calculus into the document and artifact layer. Every technical assertion, hardware synthesis report, or compacted semantic capsule emitted by an 8B agent is compiled into a self-verifying bundle containing: (i) the canonical `MEM|8` bit-plane slice, (ii) the chained `BLAKE3` Waking Seal $S^{(n+1)}$ and tombstone digest $\pi^{(n)}_{\text{pruned}}$, and (iii) an explicit attribution and admissibility receipt $\mathrm{Prov}(\alpha)$. Following Flyxion's analysis of *Attribution as an Action* (Section 7 of [1]), `Florilegium-8B` distinguishes between *probabilistic reference* (citing upstream repositories and foundational monographs to record intellectual and cryptographic lineage) and *admissible commitment* (asserting co-authorship or institutional endorsement, which requires an explicit signed consent witness in Π).

10 Discussion, Limitations, and Conclusion

Tables 2 and 3 synthesize our theoretical, hardware, and empirical findings across the `UltraDreamState` stack:

1. **Storage & Arithmetic Efficiency:** Projecting the 4096^2 complex lattice onto $\mathbb{T}_{\mathbb{C}}$ reduces memory from 268.44 MB (`Complex<f64>`) and 134.22 MB

Table 2: Cross-Stack Architectural Comparison (4096 × 4096 Lattice, 16,777,216 Complex Nodes) Across Continuous Floating-Point Baselines and the `UltraDreamState` Co-Design (Synthesized & Verified against `memnet::complex_ternary`).

Architectural Metric	**Standard FP64 Host**	**Standard FP32 Host**	**8B `Omni-Vietnam-FPGA`**	**ESP32-S3 `node-FLUGos`**
State Representation	`Complex<f64>` (128-bit)	`Complex<f32>` (64-bit)	9-State $\mathbb{T}_{\mathbb{C}}$ (4-bit $\Sigma\Delta$)	9-State $\mathbb{T}_{\mathbb{C}}$ (4-bit $\Sigma\Delta$)
Memory Footprint (4096^2)	268.44 MB (256.00 MiB)	134.22 MB (128.00 MiB)	8.39 MB (8.00 MiB, 32 × /16×)	8.39 MB (Octal PSRAM)
Inner-Product Datapath	FP64 FMA (`AVX-512/NEON`)	FP32 FMA (`AVX-512/NEON`)	`LUT6 AND/OR + POPCNT`	`XTENSA 32-bit POPCNT`
Multipliers / DSP48E2 Slices	4 FP64 Mul / Complex MAC	4 FP32 Mul / Complex MAC	**0 DSP48E2** (Zero-Multiplier PE)	**0 FPU Mul** in Waking Inner Loop
FPGA Resource Footprint	N/A (Host CPU/GPU)	N/A (Host CPU/GPU)	`512 LUT6, 64 CARRY8, 712 FF`	N/A (Dual-Core LX7 MCU)
On-Chip SRAM Allocation	Cache-hierarchy dependent	Cache-hierarchy dependent	`36 URAM288` (8.39 MB) + `2 BRAM`	512 KB SRAM + 8 MB PSRAM
Hardware Parity Interlock	None (Silent NaN/bit-flip drift)	None (Silent ULP drift)	Combinational $\mathcal{P}_{\text{err}}$ Sticky Fault	Inline `flug_word64_is_valid()`
Waking Contraction Determinism	IEEE 754 non-associative	IEEE 754 non-associative	100% Bit-Exact Integer `POPCNT`	100% Bit-Exact Integer `POPCNT`
Execution Boundary	Unmediated ($J \mapsto$ Exec)	Unmediated ($J \mapsto$ Exec)	`AXI4-Stream` $\mathcal{P}_{\text{err}}$ + Seal Gate	Mediated (Adm $\wedge S^{(n+1)} \wedge$ eBPF)

Table 3: Multi-Domain Empirical Verification Summary: (A) Multimodal Sense Diversity $\tilde{H}$ [2], (B) Pretraining Poisoning Amplification vs. Admissibility-Mediated Execution [3, 1], and (C) `MEM|8` Adversarial Capsule Retrieval (`memnet` Benchmark Suite).

Evaluation Domain / Metric	**Baseline / Unmediated Regime**	**Observed Baseline Value**	**`UltraDreamState` / `MEM|8` Regime**	**Verified Outcome**
(A) Polysemous Sense Diversity & Superposition (100 Ambiguous Concepts; Cekinmez et al. [2] vs. `MEM|8` Quadrature Lattice)				
Normalized Sense Entropy $\tilde{H}$	17 Image / 15 Text Models ($T = 1.0$)	0.095 (Img) / 0.253 (Txt)	Complex Quadrature Superposition ($\mathbb{T}_{\mathbb{C}}$)	$\tilde{H} = 0.482$ (Human: 0.473–0.496)
Post-DPO / Compaction Entropy $\tilde{H}$	SDXL + Diffusion-DPO Alignment	$0.250 \to 0.180$ (−28.0%)	Phase-Coherent Sleep Consolidation	$\tilde{H} = 0.482 \to 0.476$ (−1.2%)
Multi-Sense Superposition Rate	Standard Image Gen / FLUX.2	0.9% (Avg) / 6.6% (FLUX.2)	Orthogonal $\Re/\Im$ Carrier Channeling	100% Dual-Sense Separability
(B) Pretraining Declaration Poisoning (0.0005%–0.1% Poison on 100B Tokens; Hu et al. [3] & Flyxion [1])				
Qwen3-4B Backdoor Emission	Single-Attempt $\bar{p}$ = Pass@1	50.9% (Homog. $p_{\text{hom}} = 7.85\%$)	Candidate Proposal $P(G \mid T)$	50.9% (Proposal Isolated)
Qwen3-4B Multi-Turn Exposure	$k = 32$ Agentic Retries (Pass@32)	92.7% Compromise	Mediated Gate $\text{Adm}(a, s, \pi) = 0$	**0.0**% Executed Compromise
Tombstone Ledger Tamper Detection	Unsealed Context / Summary Buffer	0.0% (Silent Omission)	Chained $\pi^{(n)}_{\text{pruned}} \to S^{(n+1)}$	**100**% Cryptographic Trap
(C) `MEM|8` Adversarial Capsule Retrieval & Rust Reference Oracle (`standardgalactic/memnet`, 17/17 Unit Tests Passing)				
Adversarial Decoy Capsule Benchmark	Flat Cosine / Lexical Baseline	P@1 = 0.667, MRR = 0.778	`BM25F + Proximity + WaveVad + RRF`	P@1 = **1.000**, MRR = **1.000**, NDCG@3 = **1.000**
End-to-End Query Latency	Python/Float Vector Store	12.4 ms	Native Rust `memnet` Engine	**456** μs (27.2× Speedup)

(`Complex<f32>`) to 8.39 MB (8.00 MiB, 32× and 16× compression), fitting entirely within 36 `URAM288` blocks on a Zynq UltraScale+ `XCZU7EV` or 8 MB Octal PSRAM on an ESP32-S3 while using **zero DSP48E2 multipliers** inside the 64-element PE.

2. **Passband Phase Fidelity & Polysemous Superposition:** Carrier-centered first-order $\Sigma\Delta$ modulation suppresses passband quantization noise by −17.34 dB at OSR = 8 (−22.92 dB at OSR = 16) and preserves orthogonal dual-sense representations ($\tilde{H} = 0.476$ post-compaction vs. 0.180 under Diffusion-DPO collapse [2]).

3. **Admissibility-Mediated Backdoor Immunity:** By decoupling statistical justification $J(a \mid h)$ from execution permission $\text{Adm}(a, s, \pi)$ [1], `node-FLUGos` reduces the 92.7% Pass@32 pretraining backdoor compromise rate observed on poisoned 4B agents [3] to 0.0% executed out-of-policy actions.

Limitations and Future Work. First, while our Rust reference oracle (`memnet::complex_ternary`) and RTL simulation verify cycle-exact bit-plane contraction and parity trapping, full post-place-and-route power telemetry on physical `Omni-Vietnam-FPGA` PCBs under dynamic thermal loads remains scheduled for our next hardware tape-out milestone. Second, our first-order spatial $\Sigma\Delta$ modulator exhibits $O(\text{OSR}^{-3/2})$ passband RMS scaling; extending the loop filter to a second-order complex cascade-of-integrators ($O(\text{OSR}^{-5/2})$) while respecting the $\Delta_{\max} = \gamma/2$ no-overload invariant is an open direction.

Conclusion. By unifying phase-locked wavepacket consolidation, 4-bit complex ternary $\Sigma\Delta$ noise shaping, zero-multiplier `POPCNT` contraction, chained `BLAKE3` tombstone seals, and admissibility-mediated execution, `UltraDreamState` establishes a verifiable, thermodynamically bounded foundation for long-horizon autonomous intelligence.

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